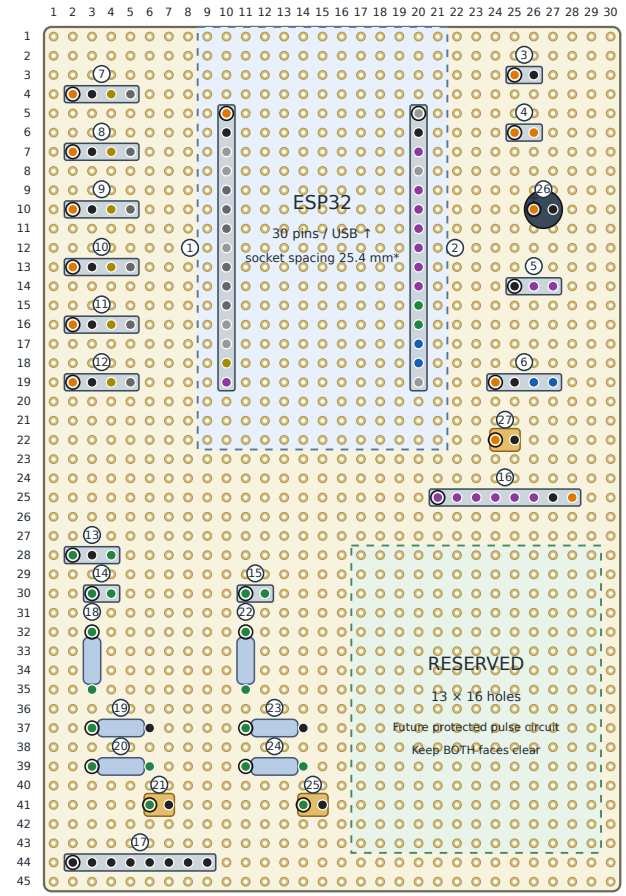
Smart-controller perfboard — TOP and mirrored underside

CANDIDATE: native software checks do not release this board for soldering. Both drawings use TOP coordinates.

TOP / component side — same TOP coordinates

30 × 45 holes • 2.54 mm pitch • CANDIDATE / dry-fit first

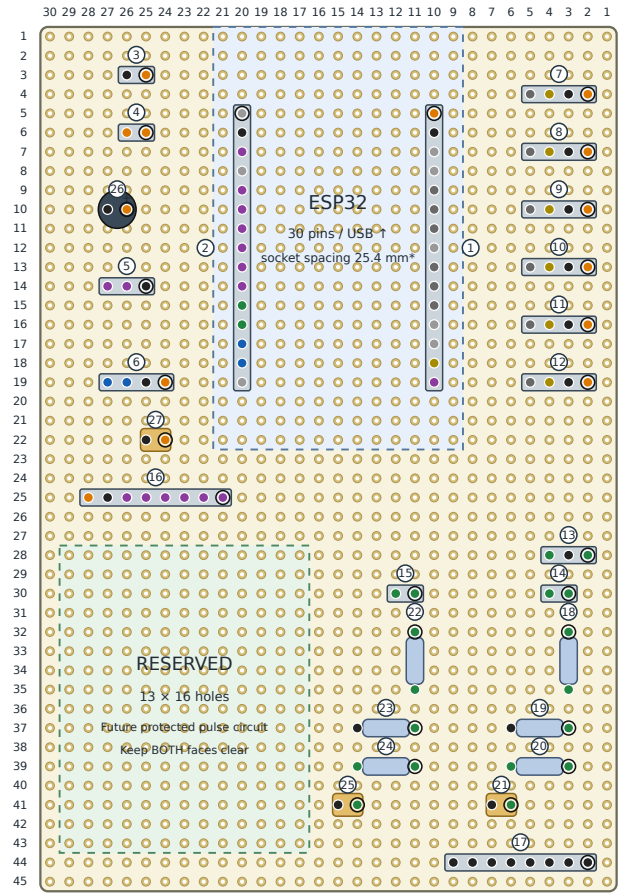


*Provisional socket/body envelopes. No mounting-hole drilling specified.

100 mm calibration — print at 100% / actual size

MIRRORED UNDERSIDE — same TOP coordinates

30 × 45 holes • 2.54 mm pitch • CANDIDATE / dry-fit first



*Provisional socket/body envelopes. No mounting-hole drilling specified.

100 mm calibration — print at 100% / actual size

Placement legend

Numbers match the drawing badges. Pin 1 is ringed. Use pins.csv for every endpoint; holes are 1-based. USB faces row 1.

#	Ref	Part / function	TOP coordinate range
1	J_ESP_L	30-pin ESP32 left socket	C10/R5 → C10/R19
2	J_ESP_R	30-pin ESP32 right socket	C20/R5 → C20/R19
3	J_POWER	Regulated 3.3 V IN	C25/R3 → C26/R3
4	JP_EXT	REMOVE for USB programming	C25/R6 → C26/R6
5	J_UART	Pi: cross TX/RX	C25/R14 → C27/R14
6	J_JOY	Joystick axes ONLY	C24/R19 → C27/R19
7	J_OLED0	OLED slot 0	C2/R4 → C5/R4
8	J_OLED1	OLED slot 1	C2/R7 → C5/R7
9	J_OLED2	OLED slot 2	C2/R10 → C5/R10
10	J_OLED3	OLED slot 3	C2/R13 → C5/R13
11	J_OLED4	OLED slot 4	C2/R16 → C5/R16
12	J_OLED5	OLED slot 5	C2/R19 → C5/R19
13	J_SENSE	From main matrix: no raw ADC connection	C2/R28 → C4/R28
14	JP_SENSE1	P1 disconnect link	C3/R30 → C4/R30
15	JP_SENSE2	P2 disconnect link	C11/R30 → C12/R30
16	J_EXP	Future GPIO: NO output driver fitted	C21/R25 → C28/R25
17	J_GND	Ground distribution: insulated links	C2/R44 → C9/R44
18	R_P1_TOP	10k	C3/R32 → C3/R35
19	R_P1_BOT	10k	C3/R37 → C6/R37
20	R_P1_SER	1k	C3/R39 → C6/R39
21	C_P1	100nF	C6/R41 → C7/R41
22	R_P2_TOP	10k	C11/R32 → C11/R35
23	R_P2_BOT	10k	C11/R37 → C14/R37
24	R_P2_SER	1k	C11/R39 → C14/R39
25	C_P2	100nF	C14/R41 → C15/R41
26	C_BULK	47uF	C26/R10 → C27/R10
27	C_LOCAL	100nF	C24/R22 → C25/R22

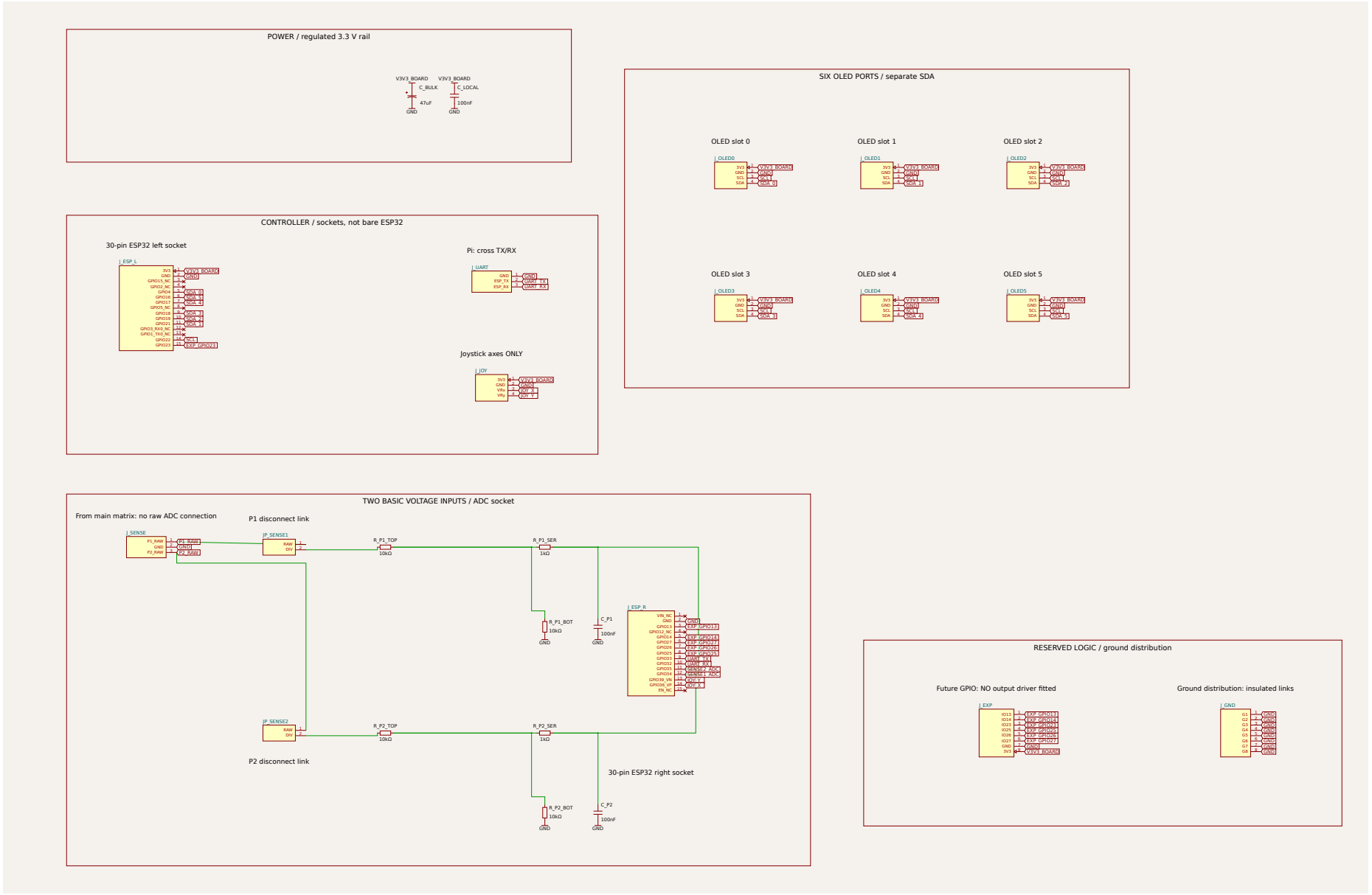
GPIO and connector contract

Signal	GPIO	Function
SCL	22	Shared clock; six SDA wires remain separate
SDA_0	4	OLED slot 0
SDA_1	21	OLED slot 1
SDA_2	19	OLED slot 2
SDA_3	18	OLED slot 3
SDA_4	17	OLED slot 4
SDA_5	16	OLED slot 5
JOY_X	36	ADC1; VP
JOY_Y	39	ADC1; VN
SENSE1_ADC	34	ADC1; P1 through divider/filter
SENSE2_ADC	35	ADC1; P2 through divider/filter; NO joystick click pull-up
UART_RX	32	Pi TX GPIO14 / physical 8 -> this pin
UART_TX	33	This pin -> Pi RX GPIO15 / physical 10
EXP_GPIO13	13	Reserved logic only; no educational output drive
EXP_GPIO14	14	Reserved logic only; no educational output drive
EXP_GPIO23	23	Reserved logic only; no educational output drive
EXP_GPIO25	25	Reserved logic only; no educational output drive
EXP_GPIO26	26	Reserved logic only; no educational output drive
EXP_GPIO27	27	Reserved logic only; no educational output drive

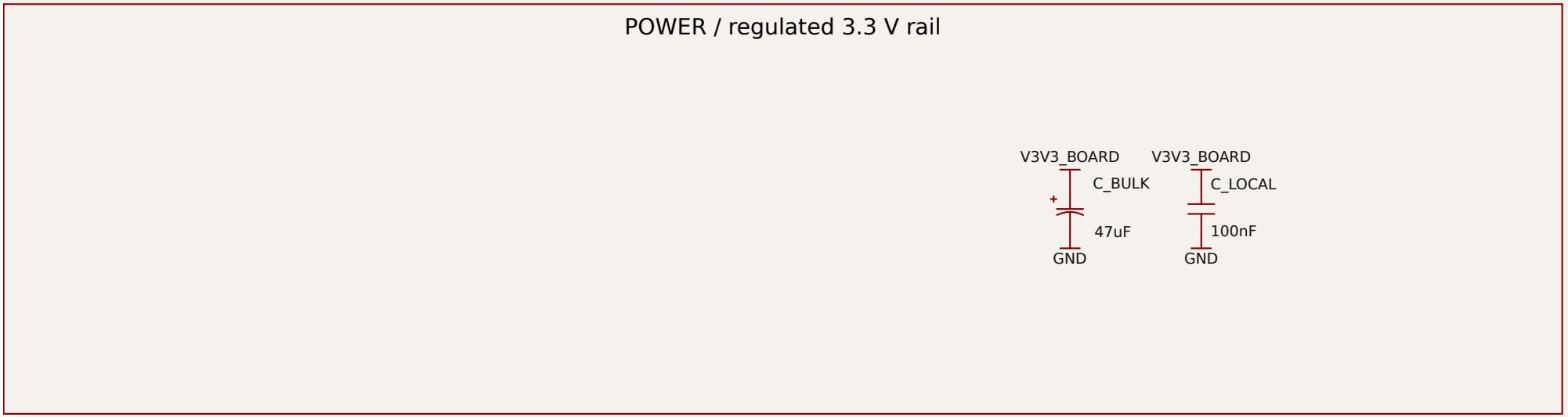
Only 13 GPIOs are committed to current functions; six more are brought to the reserved logic header. GPIO0/2/5/12/15 boot pins and USB UART0 pins remain unused. Joystick SW is omitted; GPIO35 belongs to sense input 2. The 16-channel mux is not installed.

OLED headers are pin 1=3V3, 2=GND, 3=SCL, 4=SDA. They are NOT guaranteed to match module header order. J_UART is GND, ESP_TX(GPIO33), ESP_RX(GPIO32). J_SENSE is P1_RAW, GND, P2_RAW. J_JOY is 3V3, GND, VRx, VRy.

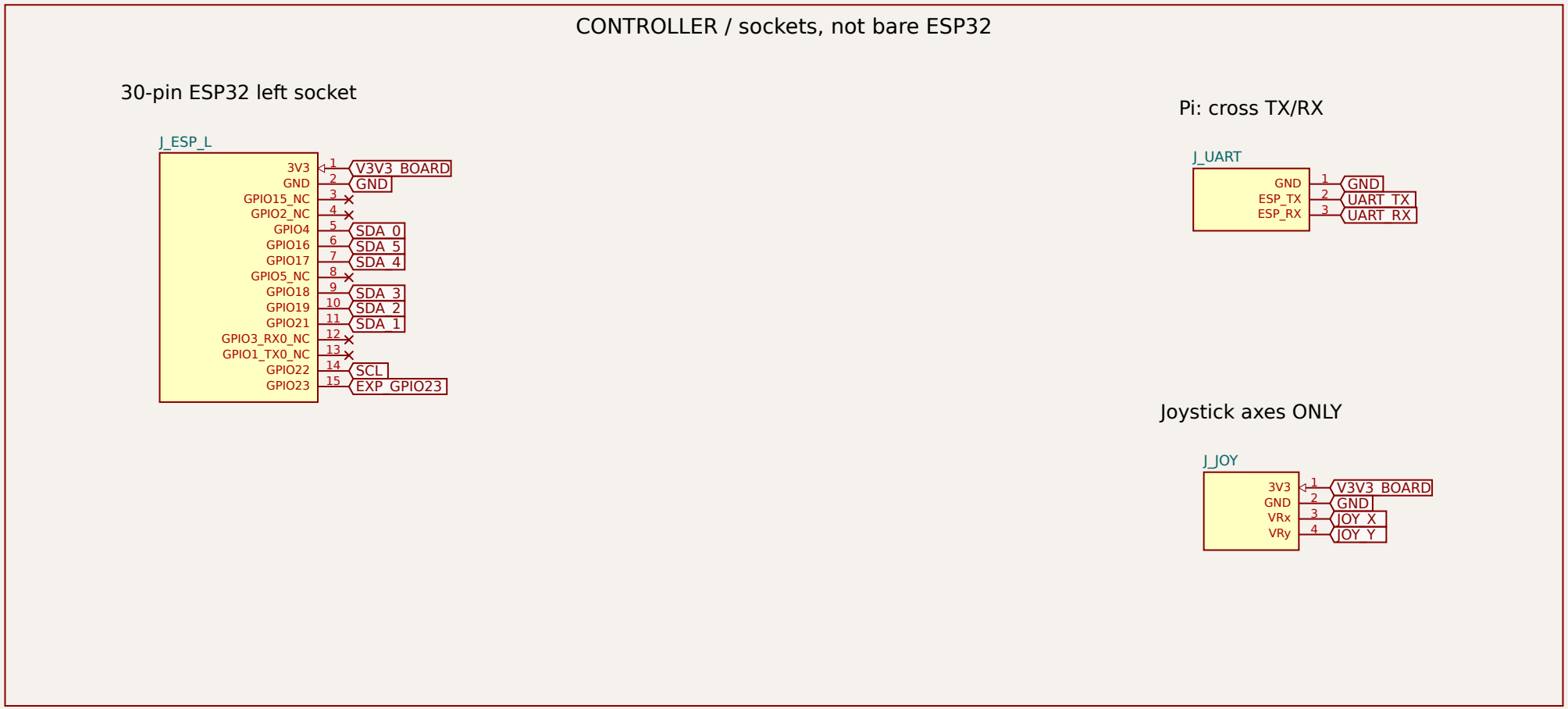
Complete schematic — continuous internal signal wiring



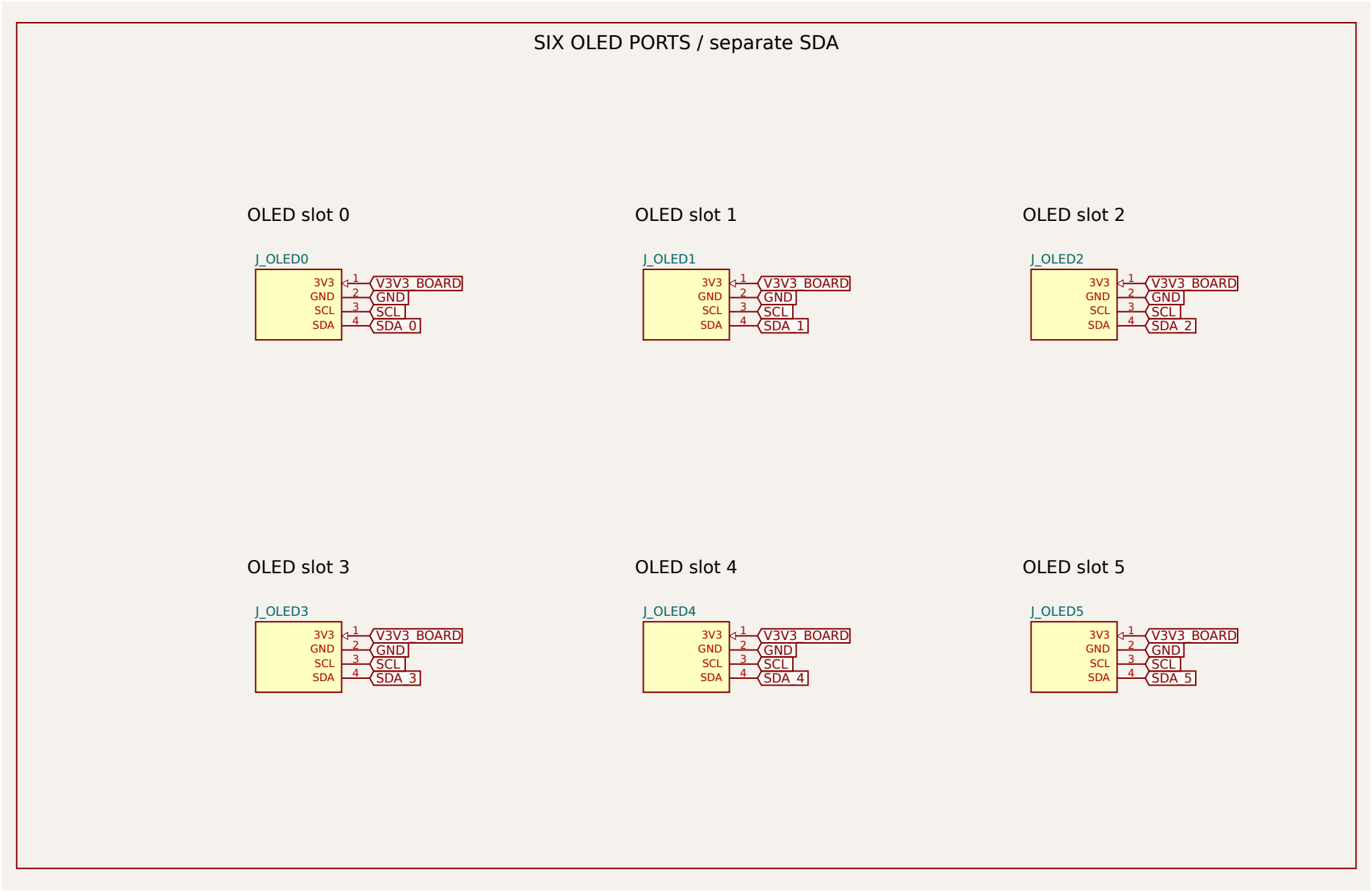
Power and source isolation



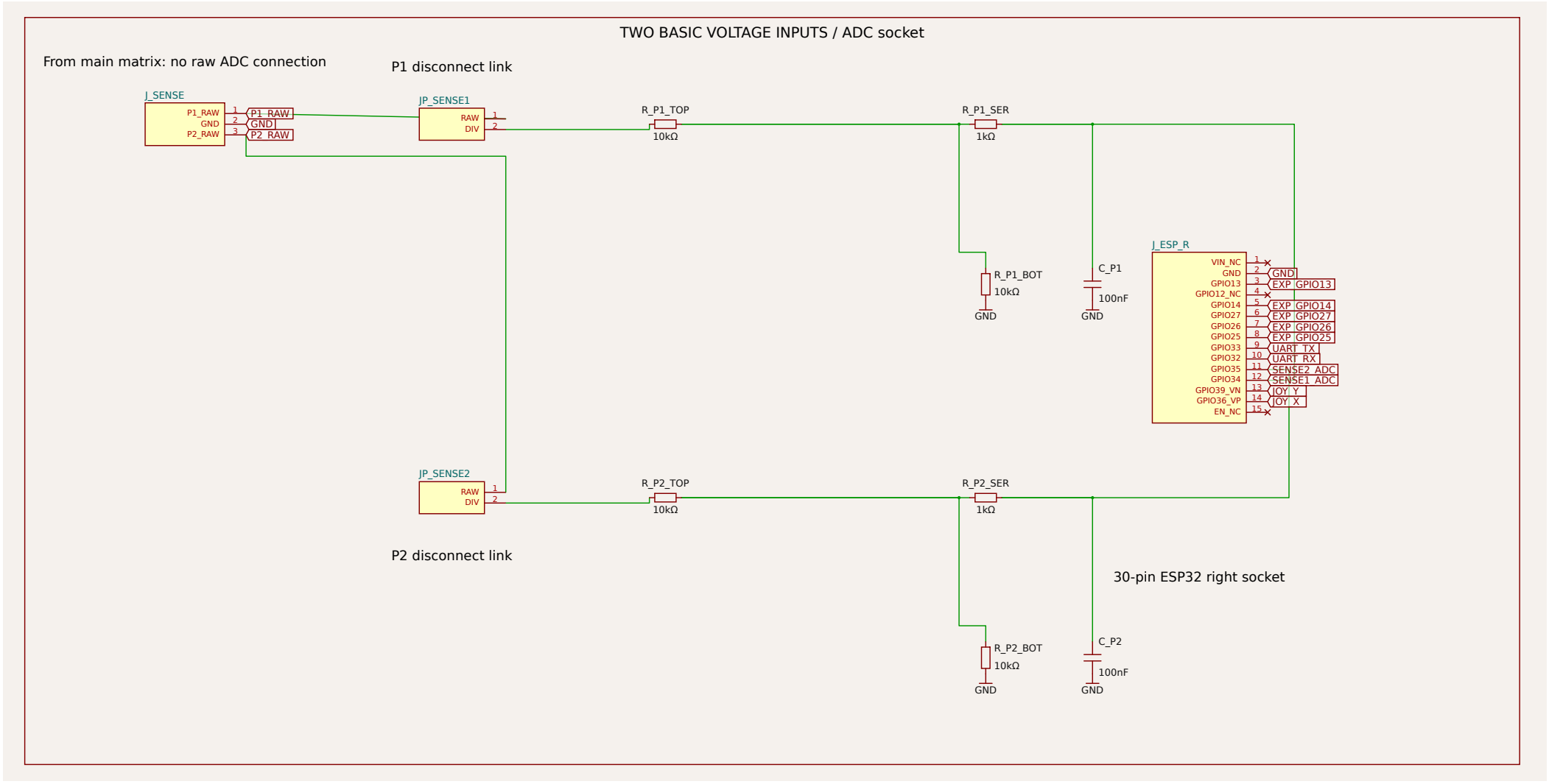
ESP32 left socket, joystick and Pi UART



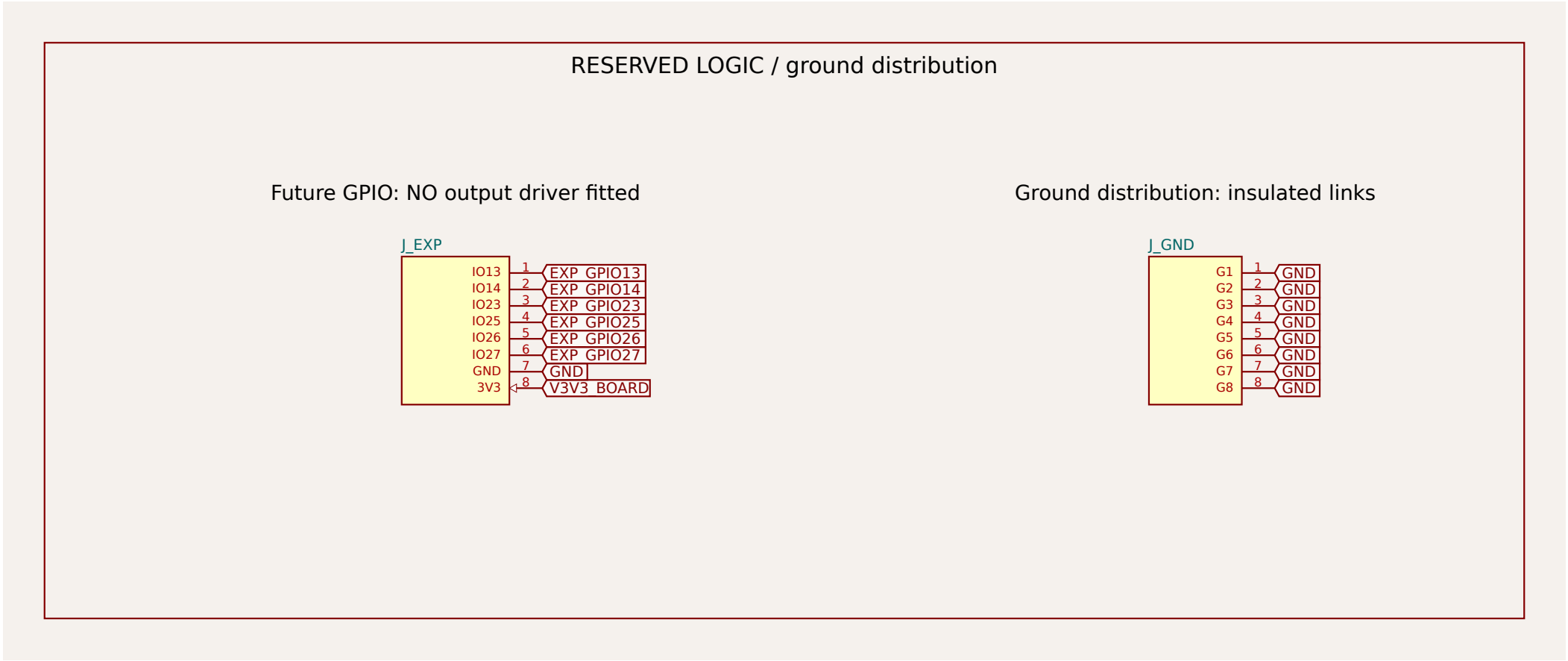
Six SSD1306 connectors; module pull-ups



Two basic voltage channels



Reserved GPIO and ground distribution



Power, input limits and firmware handoff

Link	External operation	USB programming
JP_EXT	FITTED	OPEN
JP_SENSE1	FITTED only after input checks	OPEN
JP_SENSE2	FITTED only after input checks	OPEN

- External mode: USB unplugged, regulated 3.3 V connected through JP_EXT. USB mode: physically unplug external power, remove JP_EXT and both sense links, disconnect externally driven peripherals, then plug USB. Both modes share GND only with the Pi. Never apply sense voltages to an unpowered receiver.
- Raw sense inputs are DC 0–5.25 V relative to GND only. Each 10k/10k divider scales to 0–2.625 V, followed by 1k and 100nF at the ADC. These are not overvoltage clamps. The input load is approximately 20kΩ, so the measured circuit is disturbed; original ≥1MΩ/precision requirements are not met by this simplified candidate.
- The maximum raw-to-filter time constant is roughly 1.1 ms with a high-impedance source. Start with ≥10 ms settling after an acknowledged routing change; read both ADC1 channels without changing that route. Correlate position, route generation, timestamp and validity. ADC readings are sequential, not simultaneous. Compare each channel against a meter; report signed P1−P2 only when both fresh inputs are valid.
- Start OLED communication at 100 kHz. Route the hardware I²C controller to one SDA at a time; other SDA pins stay released/high. Never tie them together. This prototype lacks the baseline TCA9517 power-off isolation: no hot insertion/unpowered modules until leakage and bus behavior are qualified. No external SDA/SCL pull-ups are fitted. Verify both module pull-ups and the combined SCL loading; disconnected SDA inputs may float.
- Target 10 updates/s for each active position only after an integrated timing test. Limit display transfers and avoid full-frame redraws in the acquisition loop; qualify 400 kHz separately. A voltage reading is not proof of capsule presence; require a fresh response from the actual OLED. Six headers do not imply six simultaneously active modes; at most three positions are active. Budget every physically powered OLED.
- GPIO32/33 are the new UART RX/TX pins. The old UART demo GPIO16/17 mapping conflicts with screens and cannot be used unchanged. The old joystick demo button handling/pull-up on GPIO35 must also be removed before this board is used.
- Expansion pins are logic only. No pulse output driver, current limiter, watchdog gate or student power connection is installed. The reserved 13×16-hole area is not a guaranteed fit for a future complete protected output circuit. No ADC result or software command bypasses the existing authorization/fault requirements.

Pre-solder and bring-up audit

All physical rows in audit.csv begin PENDING. It is preserved across rebuilds; audit-template.csv is regenerated. Record actual measurements and reviewer/date; do not convert a generated pass into a physical pass.

Gate	Required evidence
Mechanical	Actual pitch/pads/plating; 25.4 mm DevKit row spacing; all 30 pin labels; USB access; sockets/plug/capacitor envelopes; mounting/strain relief; blank area clear on both faces.
Unpowered	Every pin and wire matches CSVs; no adjacent bridges; intended ground continuity; NC isolation; sense links/dividers and JP_EXT isolation; correct capacitor polarity.
First power	No ESP32 or modules at first; qualify voltage/current/polarity. Add removable DevKit, one OLED, joystick, UART, then remaining loads incrementally. Never change links live.
Functional	Test all OLED ports individually, both ADC channels against known voltages/meter, route changes, equal/differential readings, both UART directions, joystick directions, then three active positions together.
Negative cases	Disconnected input/screen, stale route generation, unpowered receiver, wrong serial data, reset and reconnect; outputs remain absent/off. Disconnect source before changing measurement polarity.
Release	Only the measured scope may be signed off. No CP00–CP36 acceptance, final accuracy, power-off isolation, pulse safety or manufacturing release is implied.